Soil moisture analysis via soil moisture analyser

Quick Overview

Parameter	Information
Protocol ID	PHYS-003
Version	1
Method Owner	Lieke Lipsch
Estimated Hands-on Time	2 minutes per sample
Total Duration	10 – 20 minutes for a sample of 2 grams of soil
Difficulty Level	☒ **Basic** ☐ Intermediate ☐ Advanced
Training Required	☒ **Yes** ☐ No
Biosafety Level	/
Typical Throughput	1
Sample types	Soil samples, plant samples, and liquids
Output	Moisture content of your sample (%)
Related Protocols	PHYS-002

Method Information

Method Title: Soil moisture analysis via soil moisture analyser

Author(s): Lieke Lipsch

Date Created: 05-06-2025

Last Updated: 07-09-2026

Reviewed By:

Approved By:

Purpose

This method is used to determine the moisture in (soil) samples.

Scope

Suitable for

- Most types of samples including soils, plants, and liquids, as long as no corrosive and/ or toxic chemicals are present.
- The weight range of samples that can be measured is: 0.5 g – 71 g (see notes about the standard deviation at point 4 below).

Not suitable for

- Old/ dried out samples
- Any toxic or corrosive chemicals, as this can lead to toxic fumes.

Safety Considerations

Hazards

- Biological: /
- Chemical: Do not put in any toxic or corrosive chemicals, as this can lead to toxic fumes.
- Physical:

- During and after use, parts of the instrument will be hot.
- Do not touch the areas marked with warning symbols.
- Do not open the device before it has cooled down.

Required PPE

- Lab coat
- Gloves

Waste Disposal

- Biological waste: Contaminated soils should be disposed of appropriatly
- Chemical waste: /
- Sharps:/

Materials and Reagents

Reagent/Material	Specification	Supplier	Storage
Sample pans	Aluminum sample pan, HA-D90, set of 80 pcs 00013865	https://www.mt.com/shop/gb/en/product/13865/aluminium-sample-pans	In drawer underneath the moisture analyser in the soil lab 205

Equipment

Equipment	Model	Location
Moisture analyser	HE73- METTLER TOLEDO	205

Procedure

Before Starting

Checklist

☐ Required training completed

☐ Equipment available

☐ Sample IDs assigned

Step-by-Step Procedure

General information

- The standard program will heat your sample to 105 degrees Celsius and will stop the measurement when your sample is completely dry i.e. there is no weight loss measured anymore. There are other options, see tips at the bottom for this.
- When using the standard program described above, it takes approximately 10-15 minutes for a sample of 2 grams of soil.
- One sample can be measured at a time. After a short cooling period and removing the previous sample the next sample measurement can be started.
- You will get the most accurate reading if the device has been plugged in for at least one hour.

How to use the moisture analyser:

1. You can turn on the device by pressing the 'on' button. Check if the device is leveled correctly, by making sure that the air bubble is in the center of the circle.
2. Blinking icons on the display will show which step you need to take at that moment, as shown below (blinking parts indicated in gray):
3. Place the sample pan in the holder and make sure that it is level. It is important that the sample pan is not bend, use a new one if this is the case. New sample pans are stored in a blue box in the top drawer below the device. Close the lid to tare (no need to press any buttons, it will tare automatically).

	Basic weighing	Load the empty sample pan and execute a tare. **Note** Tare is only possible with closed heating module.

4. You can now place your sample in the sample pan. Make sure that your sample is spread out evenly over the bottom of the sample pan. The standard deviation from larger samples is smaller compared to small samples (max. 0.05% for 2g samples, max. 0.01% for 10g samples). So it is likely best to use at least 2 g of each sample. The required minimum weight of a sample is 0.5g (maximum 71 g).

	Ready for weighing	Place the sample on the sample pan.

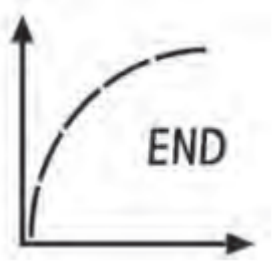
5. Close the lid.

	Close heating module	Close heating module.

6. Drying will start automatically.

	Ready for start	Start the drying process.

7. The device will beep as soon as the drying is finished. 'END' will be showing in the progress indicator diagram. The display will indicate the moisture content (MC) of your sample in %.



8. After cooling you can open the lid by using the sides (indicated here by the black arrows). Do not use the gray handle of the sample tray holder; you will spill your sample.



9. Take out the sample pan by lifting the sample pan holder. You can take the pan out of the holder by lifting it up slightly. Clean the sample tray with a tissue. If any sample spilled onto the device make sure to clean this too, but be careful some parts might still be hot.
10. You can now measure another sample by repeating the previous steps.

Tips:

- Your sample will dry quicker if it has a larger surface. You can use glass fiber filters to do this for liquids.
- If you do need to measure samples that could produce toxic gases, use the device in a fume hood.
- There are other options such as programming a second type of program. This can include stopping the drying process after a certain amount of time. The device will calculate the moisture using the weight lost within this time. This can be useful to make the analyses shorter. Please confer the handbook, that can be found here: https://www.hogentogler.com/images/Mettler_HE73_manual.pdf?srsltid=AfmBOopjCxtCQHZcgqIXAZFmqFY_fN1nG0YPj9t_6cjIsb26p7YJp48s

Calculation:

Calculation used by device for the calculation of the moisture content:

$$MC = \frac{WW - DW}{WW} \cdot 100\ \%$$

MC = Moisture Content [0…100 %]
WW = wet weight
DW = dry weight

Critical Control Points

⚠ Step: Balancing the sample pan

Why it is critical: If the pan is not level no accurate measurements can be taken.

How to avoid mistakes: Make sure that the pan sit horizontally in the holder and is touching all three arms of the holder.

Quality Control

QC Parameter	Acceptance Criterion

Troubleshooting

Problem	Possible Cause	Solution

Expected Outcome

The moisture content of your sample can range between 0-100%.

Data Recording in Lab Books

Record the following information:

- Operator
- Date
- Sample IDs
- Batch numbers
- Deviations from protocol
- Raw data location

Storage and Stability

Item	Storage Condition	Stability

Soil samples	Samples should be stored in closed vials or bags to prevent excessive drying.	Soil samples should not be stored long before measurement

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References

- SOPs
- Publications
- Instrument manuals

Revision History

Version	Date	Changes	Author
1.0	07-09-2026	Initial version	Lieke Lipsch